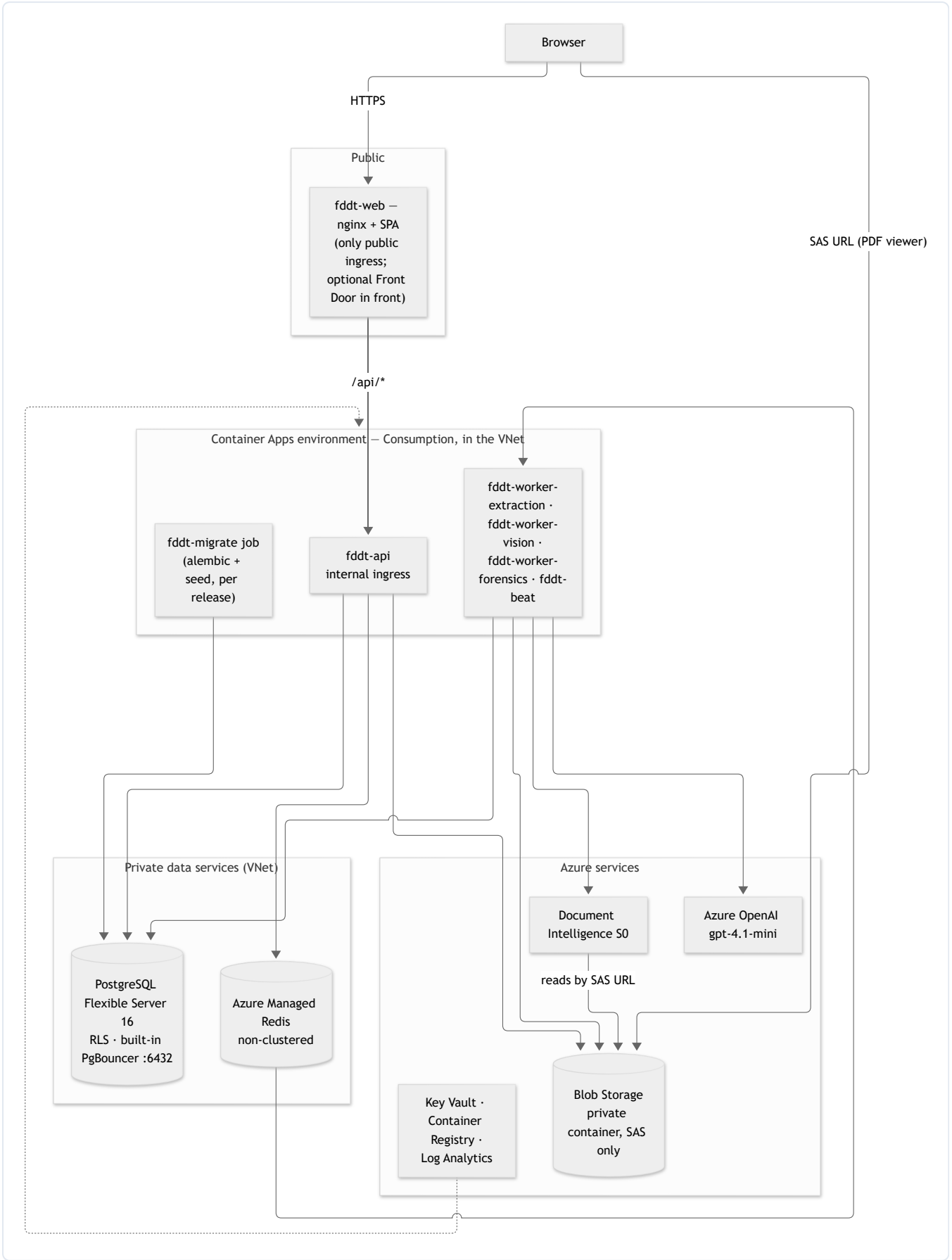
Production deployment (Azure)

> **Status:** this is the deployment plan and runbook. Per the project rules no infrastructure-as-code or CI/CD is committed yet — the repository ships the two container images (`backend/Dockerfile`, `frontend/Dockerfile` + `nginx.conf.template`) and everything below is done with the Azure CLI or portal. Prices are **list prices in USD for UAE North, retrieved 2026-10-03** from the Azure Retail Prices API; they exclude VAT and any EA/CSP discount. Re-check them in the pricing calculator before quoting.
>
> **To deploy, follow azure-production-runbook.md**, the complete copy-paste runbook (network/IP/port plan, NSGs, DNS, YAML for every app, autoscaling, alerts, release and rollback). Section 4 below is the condensed version of it.

1. Target architecture



Component	Azure service	Why
SPA + reverse proxy	Container App `fddt-web` (frontend image), **public** ingress	One origin for the SPA and `/api`, so no CORS is needed; nginx caps request bodies (`UPLOAD_BODY_LIMIT` / `BULK_UPLOAD_BODY_LIMIT`, at least the largest per-company limit)
API	Container App `fddt-api` (backend image), **internal** ingress only	Never exposed directly; reached only through `fddt-web`
Workers	Container Apps `fddt-worker-extraction`, `-vision`, `-forensics` (backend image), no ingress	One per queue so each scales on its own (processing-queues.md)
Scheduler	Container App `fddt-beat`, no ingress, **min = max = 1**	Beat plus a one-slot worker for `housekeeping_queue`: per-minute queue metrics, stuck-document recovery every 5 minutes, nightly usage reconciliation. Two beats would run every job twice
Migrations	Container Apps **job** `fddt-migrate` (manual trigger)	Runs `alembic upgrade head` + `seed.py` once per release, as the database owner — not on every API start
Database	Azure Database for PostgreSQL **Flexible Server 16**, private access (VNet)	Row-Level Security, roles without superuser/BYPASSRLS (verified to work without superuser); built-in PgBouncer on General Purpose tiers
Broker / limiter / fair-share	**Azure Managed Redis**, *Non-Clustered* policy	Azure Cache for Redis no longer accepts new customers (since 2026-04-01) and retires 2028-09-30
Files	Storage account (Standard LRS/ZRS, hot), one **private** container	Originals (write-once), signature crops, report PDFs; accessed only through short-lived SAS URLs
OCR	Document Intelligence **S0**	`prebuilt-layout`
LLM	Azure OpenAI / Foundry, **gpt-4.1-mini** deployment	Classification, extraction, visual review, signatures, issuer match
Secrets	Key Vault, referenced as Container Apps secrets	No secret in images or the repo
Images	Container Registry (Basic)	Backend and frontend images
Observability	Log Analytics workspace (Container Apps logs) + Azure Monitor alerts	`queue_alert`, failed documents, database metrics
Edge (optional, not used by default)	Front Door Standard	Global edge, caching of static assets, managed certificate. Container Apps ingress already provides TLS and a free managed certificate, and sign-in throttling is built into the app, so this is optional. WAF custom rules work on Standard; the managed OWASP rule sets need **Front Door Premium** ($330/month base)

Region and data residency

- **UAE North** is used for everything that *stores* data (Storage, PostgreSQL, Redis, Document Intelligence processing). The documents are mostly UAE/Saudi.
- **Azure OpenAI gpt-4.1-mini is only offered as a *Global* deployment in UAE North** (no regional or Data Zone SKU there). With a Global deployment, prompts — the OCR text and page images — may be **processed in any**

Azure region; data at rest stays in the resource's region. If the client's data classification forbids that, use a region/model combination that offers a regional deployment, or have the client accept Global processing in writing. (In West Europe, a Data Zone EU deployment costs 10 % more: $0.44 / $1.76 per 1M tokens.)

- Review the Azure OpenAI data-handling and abuse-monitoring terms against the client's policy; request modified abuse monitoring if required.

2. Service tiers

Three sizes. All three use the same images and the same configuration keys — only replica counts, CPU and SKUs differ, so moving between them is a scaling change, not a redeployment.

	S — Pilot	**M — Production**	**L — Scale**
Volume (2-page documents/month)	≈ 3,000	≈ 20,000	≈ 100,000
Companies	1–3	up to ~20	20+
`fddt-web`	1 × 0.25 vCPU / 0.5 GiB	2 × 0.25 / 0.5 (max 4)	2 × 0.25 / 0.5 (max 6)
`fddt-api`	1 × 0.5 / 1 (max 2)	**1** × 1 / 2 (HTTP autoscale to 4)	3 × 1 / 2 (max 6)
`fddt-worker-extraction`	1 × 0.5 / 1	1 × 1 / 2	2 × 1 / 2
`fddt-worker-vision`	1 × 0.5 / 1	1 × 1 / 2	4 × 1 / 2
`fddt-worker-forensics`	**0**–2 × 1 / 2 (scale to zero)	**0**–4 × 2 / 4 (scale to zero)	1–10 × 2 / 4
`fddt-beat` (beat + housekeeping worker)	1 × 0.5 / 1	1 × 0.5 / 1	1 × 0.5 / 1
PostgreSQL	Burstable **B2s** (2 vCore, 4 GiB), 64 GB, no HA, direct connections	General Purpose **D2ds_v5** (2 vCore, 8 GiB), 128 GB, **no HA at launch**, built-in PgBouncer	General Purpose **D4ds_v5** (4 vCore, 16 GiB), 512 GB, zone-redundant HA, built-in PgBouncer
Redis	Managed Redis **B0**, no HA	Managed Redis **B0**, **no HA**	Managed Redis **B3**, HA
Azure OpenAI gpt-4.1-mini Global Standard TPM (target; no quota is requested, the deployment uses the subscription's free capacity)	100k	150k	500k
Availability	single zone, no failover	single zone (99.9 % database SLA; a zone failure means a restore, ~1 hour); 2 web replicas	zone-redundant database + Redis (99.99 %)

Notes on sizing:

- **High availability (HA) on tier M is a client decision.** Without it the database is one server in one zone: Azure restarts it after a crash, and a whole-zone outage means a point-in-time restore (about an hour). Turning HA on later is one command and roughly doubles the database cost (+$176/month): `az postgres flexible-server update -g $RG -n $PG --high-availability ZoneRedundant`. Record the decision with the client.
- **Redis without HA is safe** because Redis holds no durable data, and the stuck-document job (processing-queues.md) re-queues any document whose tasks were lost when Redis restarted. Tier L keeps HA so that a Redis

restart doesn't pause a busy platform for minutes.

- **Forensics memory:** pages are rendered at 200 DPI in memory; give each forensics *process* ~2 GiB (`start-worker.sh` runs one prefork process per vCPU of the container's limit).
- **Forensics scales to zero** (tiers S/M): KEDA starts a replica as soon as a forensics task waits, and stops it after ~5 idle minutes. The first document after a quiet period waits ~30–60 s longer for its forensic checks. This works because the every-minute platform jobs run on their own `housekeeping_queue` in the beat container, so nothing else wakes the forensics workers.
- **Vision/extraction** pools are I/O-bound threads; their real ceiling is the Azure OpenAI quota. Throughput ≈ (TPM cap) ÷ ~32k tokens *reserved* per 2-page document — e.g. 80k TPM ≈ 2.5 documents/min, 400k TPM ≈ 12/min. No quota is requested: the deployment uses the subscription's free capacity, which is then the ceiling (processing-queues.md). They keep one replica each (no scale-to-zero), so OCR of a new document starts at once; more replicas than the quota can feed don't help.
- **The beat container** runs beat plus a one-slot worker for `housekeeping_queue` (queue metrics every minute, stuck-document recovery every 5 minutes, nightly usage reconciliation). Always exactly one.

3. Costs

Unit prices used (UAE North, USD, list)

Service	Meter	Price
Container Apps (Consumption)	vCPU active / idle	$0.000024 / $0.000003 per vCPU-second (≈ $63.07 / $7.88 per vCPU-month)
	Memory (**same rate active or idle**: there is no idle discount on memory)	$0.000003 per GiB-second (≈ $7.88 per GiB-month)
	Free grant per subscription per month	180,000 vCPU-s + 360,000 GiB-s + 2M requests (≈ $5.40)
	Environment management (only if environment private endpoint or planned maintenance is enabled)	$0.143/hour (≈ $104/month) — **not used** in this design
PostgreSQL Flexible Server	Burstable B2s	$0.079/hour
	General Purpose Ddsv5	$0.1085 per vCore-hour (D2ds_v5 $0.217, D4ds_v5 $0.434); 1-year reserved $571 per vCore-year (≈ 40 % less), 3-year $1,141 per vCore (≈ 60 % less)
	Storage / backup storage beyond the free 100 %	$0.138 / $0.105 per GB-month (HA doubles compute **and** storage)
Azure Managed Redis (Balanced)	B0 / B3 per instance	$0.0726 / $0.09295 per hour (HA = 2 instances). *The UAE North AMR meters look inconsistent (B1 is listed above B3; West Europe lists B0 at $0.019) — confirm in the calculator; the figures here are the higher, conservative ones*
Blob Storage	Hot LRS data / writes / reads	$0.0203 per GB-month / $0.06 per 10k / $0.0048 per 10k
Document Intelligence S0	Layout (pre-built) pages	**$10 per 1,000 pages**; commitment tiers from 20k pages for $190/month (overage $9.50/1k) and 100k pages for $900/month ($9/1k)

Service	Meter	Price
	Font-style add-on (`styleFont` , requested on every page: `AZURE_DOCUMENT_INTELLIGENCE_STYLE_FONT=true`)	**≈ $6 per 1,000 pages — to confirm** on the Document Intelligence pricing page (add-on capabilities). It feeds the font consistency check on scanned pages
Azure OpenAI gpt-4.1-mini (Global)	Input / cached input / output	**$0.40 / $0.10 / $1.60 per 1M tokens**; Batch API half price (not usable for this interactive pipeline)
Container Registry	Basic	$0.1666/day (≈ $5/month)
Key Vault	Standard operations	$0.0396 per 10,000
Log Analytics	Analytics Logs ingestion (first 5 GB/month free)	$3.29/GB
Azure Monitor alerts	Log alert at 5-minute frequency	$1.50/month each
Front Door Standard (optional, not in the estimate)	Base fee + requests/data	$35/month + usage
NAT Gateway (optional, not in the estimate)	Gateway + static IP + data	≈ $33/month + $0.045/GB
Azure Bastion Standard (on demand only, not in the estimate)	Per hour while it exists	≈ $0.29/hour (≈ $210/month if left running)
Bandwidth	Internet egress	First 100 GB/month free, then $0.075–0.12/GB

AI cost per document (measured)

Token use is taken from the app's own measurements of real calls (`fddt:llm_usage:*` , 12 English + Arabic sample documents):

Call	Per	Real input tokens	Real output tokens
Classification + extraction	document	2,105	889
Visual review (2 runs)	page	2 × 2,904	2 × 177
Signature/stamp detection	document (~3 calls)	3 × 1,418	3 × 84
Issuer semantic match	document, only if fuzzy match fails (assumed always)	400	126
Signature comparison	on demand only	1,055	47
Spelling judgment (English) — *estimate, not yet measured*	document, only if the dictionary pass leaves candidates (most clean documents: none or 1–3)	~600 + ~60 per candidate	~70 per candidate

Document size	Azure OpenAI	Document Intelligence layout	Font add-on (to confirm)	**AI cost per document**
1 page	$0.0076	$0.0100	$0.006	**$0.024**
2 pages	$0.0105	$0.0200	$0.012	**$0.043**

Document size	Azure OpenAI	Document Intelligence layout	Font add-on (to confirm)	**AI cost per document**
3 pages	$0.0134	$0.0300	$0.018	**$0.061**
5 pages	$0.0192	$0.0500	$0.030	**$0.099**

Document Intelligence (per page) is the larger part; Azure OpenAI is ~1 cent per document. The spelling judgment (added after these measurements) adds well under a tenth of a cent for a typical document, so it is not in the totals above. Without the font add-on (`AZURE_DOCUMENT_INTELLIGENCE_STYLE_FONT=false`) a 2-page document costs $0.031, but scanned pages then get no font consistency check (fonts on scans are left to the visual review).

Monthly estimate per tier (2-page documents)

	S — Pilot	**M — Production**	**L — Scale**
Container Apps (range: see below)	$75–130	$180–300	$730–1,180
PostgreSQL	$67	$176 (no HA)	$775 (HA)
Managed Redis	$53	$53 (no HA)	$136 (HA)
Blob Storage	$1	$4	$19
Container Registry, Key Vault	$6	$9	$21
Log Analytics + alerts	$10	$10–15	$25–60
Infrastructure	**≈ $210–265**	**≈ $430–555**	**≈ $1,705–2,190**
Document Intelligence layout	$60	$400	$2,000
Document Intelligence font add-on (to confirm)	$36	$240	$1,200
Azure OpenAI	$32	$210	$1,051
AI (usage-based)	**≈ $128**	**≈ $850**	**≈ $4,251**
Total per month	**≈ $340–395**	**≈ $1,280–1,405**	**≈ $5,955–6,440**
Per document	≈ $0.11–0.13	≈ $0.064–0.070	≈ $0.060–0.064

The previous design (always-active billing, HA on M, 2 forensics + 2 API replicas always on, Front Door, 30 GB of logs) estimated infrastructure at $369 / $1,283 / $2,636. The changes above cut it by 28–43 % / 57–66 % / 17–35 %. The AI line grew only because the font add-on is now counted.

How the figures were computed (730 h/month):

- **Container Apps, low end:** replicas sitting at their minimum with no work are billed the **idle** vCPU rate (Azure applies it automatically when a replica uses < 0.01 vCPU and serves no requests; a Celery worker waiting on Redis normally qualifies). Activity assumed: web/API active 30 % of the month (working hours), extraction/vision active 15 % (M) / 60 % (L) of the time, forensics billed only while it has work (scale to zero; tier L ~1 replica on average), beat idle. **Memory is always billed in full.**
- **Container Apps, high end:** the same, but web, API, extraction, vision and beat billed at the active rate all month (in case Azure does not treat the idle workers as idle).
- **Logs:** the app's own log volume is small (a few lines per task, one metrics line per queue per minute), so S/M most likely stay within the free 5 GB/month; the workspace gets a 1 GB/day cap as a guard. Alerts ~$8/month.
- Blob assumes ~0.5 MB per document retained plus ~4 writes and ~40 reads per document.
- Not included: VAT, support plans, Defender for Cloud, Front Door, NAT Gateway, Bastion, egress beyond 100 GB, signature comparisons, staging (a staging copy of tier S, paused when unused, costs well under tier S).

- **Check it after the first month** in Cost Management (filter by tag `app=fddt`), and adjust the replica sizes to what the apps really use.

Cost levers

Already applied in this design

Lever	Effect
Idle billing + forensics scale-to-zero (S/M), housekeeping jobs on their own queue	Most of the Container Apps saving
`fddt-api` minimum 1 replica on M (autoscale to 4); rolling updates still cause no downtime	≈ −$25–80/month
PostgreSQL and Redis without HA on M at launch	−$176 and −$53/month (re-enable when the client needs 99.99 %)
Sign-in throttling in the app (Redis) instead of a Front Door WAF	−$35+/month
Logs kept in the normal (Analytics) tier but small, with a 1 GB/day cap. *Basic Logs* are not used: the alert rules read the same table, and Basic tables can't drive them	−$60–200/month vs. the old estimate
NAT Gateway only if someone must allow-list our outbound IP; Bastion created only for admin sessions, then deleted	Avoids $33 and $210/month

Later, once usage is known

- **PostgreSQL 1-year reservation** after 1–2 months on a stable size: ≈ 40 % off compute (M without HA: −$63/month; L: −$254). It is a commitment, paid even if you resize or delete.
- **Document Intelligence commitment tier** once volume is steady: at 200k pages/month (tier L), the 100k-page commitment ($900) + 100k overage at $9/1k = $1,800 instead of $2,000.
- **Font add-on:** confirm its price. If the font consistency check on scanned pages is not worth it, set `AZURE_DOCUMENT_INTELLIGENCE_STYLE_FONT=false` (−$12 per 1,000 two-page documents).
- **Right-size extraction/vision** after a few weeks of real metrics: 0.5 vCPU / 1 GiB each saves ≈ $8–40/month per app.
- **Azure savings plan for compute**: check whether it covers Container Apps consumption in the client's agreement (≈ 15 % off).
- **Staging:** stop its PostgreSQL server (`az postgres flexible-server stop`; not billed for compute while stopped, up to 7 days at a time) and scale its apps to zero when unused.
- Keep pages per document honest: the cost scales with pages, and every page is OCR'd and reviewed twice.

4. Deployment steps

All commands are Azure CLI (bash). Replace `<...>` placeholders. Run them in this order the first time; later releases only repeat steps 10 and 12.

4.1 Prerequisites and naming

```
az login
az extension add --name containerapp --upgrade
az provider register --namespace Microsoft.App
az provider register --namespace Microsoft.OperationalInsights

RG=rg-fddt-prod
LOC=uaenorth
PREFIX=fddtprod                    # lowercase, globally unique where needed
az group create -n $RG -l $LOC
```

4.2 Logging

```
az monitor log-analytics workspace create -g $RG -n $PREFIX-logs -l $LOC --retention-time 30
LA_ID=$(az monitor log-analytics workspace show -g $RG -n $PREFIX-logs --query customerId -o tsv)
LA_KEY=$(az monitor log-analytics workspace get-shared-keys -g $RG -n $PREFIX-logs --query primarySharedKey
-o tsv)
```

4.3 Network

```
az network vnet create -g $RG -n $PREFIX-vnet -l $LOC --address-prefixes 10.20.0.0/16
# Container Apps (workload-profiles environment needs a /27 or larger, delegated)
az network vnet subnet create -g $RG --vnet-name $PREFIX-vnet -n aca --address-prefixes 10.20.0.0/23 \
  --delegations Microsoft.App/environments
# PostgreSQL private access (delegated)
az network vnet subnet create -g $RG --vnet-name $PREFIX-vnet -n pg --address-prefixes 10.20.2.0/24 \
  --delegations Microsoft.DBforPostgreSQL/flexibleServers
# Private endpoints (Redis)
az network vnet subnet create -g $RG --vnet-name $PREFIX-vnet -n pe --address-prefixes 10.20.3.0/24
```

4.4 Container registry and images

```
az acr create -g $RG -n ${PREFIX}acr --sku Basic
az acr build -r ${PREFIX}acr -t fddt-backend:<version> ./backend
az acr build -r ${PREFIX}acr -t fddt-web:<version> ./frontend
```

The frontend image is built with the default `VITE_API_BASE_URL=/api`; nginx proxies `/api/*` to the API.

4.5 PostgreSQL

```
az postgres flexible-server create -g $RG -n $PREFIX-pg -l $LOC --version 16 \
  --tier GeneralPurpose --sku-name Standard_D2ds_v5 --storage-size 128 \
  --vnet $PREFIX-vnet --subnet pg --private-dns-zone $PREFIX-pg.private.postgres.database.azure.com \
  --admin-user fddtowner --admin-password '<OWNER_PASSWORD>' --backup-retention 35
# Tier S instead: --tier Burstable --sku-name Standard_B2s --storage-size 64
# Tier L (or M once the client wants 99.99 %): add --high-availability ZoneRedundant

az postgres flexible-server db create -g $RG -s $PREFIX-pg -d fddt
# Built-in PgBouncer (General Purpose / Memory Optimized only — NOT available on Burstable)
az postgres flexible-server parameter set -g $RG -s $PREFIX-pg -n pgbouncer.enabled --value true
az postgres flexible-server parameter set -g $RG -s $PREFIX-pg -n metrics.pgbouncer_diagnostics --value on
```

- The admin login (`fddtowner`) is the **owner** role the migrations run as. It has CREATEROLE through `azure_pg_admin` but is not a superuser — exactly what the migrations need (they create `fddt_app` and `fddt_platform` themselves, without BYPASSRLS).
- TLS is enforced by default (`require_secure_transport = on`); always add `?sslmode=require` to `DATABASE_URL`. The app keeps that query string when it derives the two app-role URLs and the pooler URL from it.
- Built-in PgBouncer: port **6432**, transaction pooling (the default), `default_pool_size` 50 per user/database. The app sets the tenant only transaction-locally, so it is safe behind transaction pooling (`tests/test_pgbouncer_rls.py`). It authenticates against the database, so no `userlist.txt` is used in production.

4.6 Redis

```
az redisenterprise create -g $RG -n $PREFIX-redis -l $LOC --sku Balanced_B0 \
  --clustering-policy NoCluster --high-availability Disabled \
  --public-network-access Disabled --access-keys-authentication Enabled
# Tier L: --sku Balanced_B3 --high-availability Enabled
az network private-endpoint create -g $RG -n $PREFIX-redis-pe --vnet-name $PREFIX-vnet --subnet pe \
  --private-connection-resource-id $(az redisenterprise show -g $RG -n $PREFIX-redis --query id -o tsv) \
  --group-id redisEnterprise --connection-name redis
REDIS_KEY=$(az redisenterprise database list-keys -g $RG --cluster-name $PREFIX-redis --query primaryKey -o
tsv)
```

(The `az redisenterprise` command group manages Azure Managed Redis; flags vary between CLI versions. `NoCluster` exists in the current API, but some CLI versions only accept `OSSCluster` / `EnterpriseCluster` — in that case create the cache in the **portal** and pick *Non-Clustered*. **The clustering policy cannot be changed after the database is created** — choose it correctly the first time.) Requirements for Celery:

- **Clustering policy *Non-Clustered***. Celery's Redis transport and the fair-share priority lists use multi-key commands across `<queue>`, `<queue>:1…9`; the default *OSS* policy needs a cluster-aware client, and *Enterprise* can reject cross-slot multi-key commands.
- Only database **0** exists — point **both** Celery URLs at `/0` (`rediss://:<KEY>@<name>.<region>.redis.azure.net:10000/0?ssl_cert_reqs=required`; port 10000, TLS). Result keys are prefixed, so sharing the database is safe.
- Add a private DNS zone link for the private endpoint (`privatelink.redis.azure.net`) so the apps resolve it inside the VNet.

4.7 Storage

```
az storage account create -g $RG -n ${PREFIX}files -l $LOC --sku Standard_ZRS --kind StorageV2 \
  --min-tls-version TLS1_2 --allow-blob-public-access false --https-only true
az storage container create --account-name ${PREFIX}files -n documents --auth-mode login   # private
az storage cors add --account-name ${PREFIX}files --services b --methods GET HEAD OPTIONS \
  --origins https://<your-app-domain> --allowed-headers '*' --exposed-headers '*' --max-age 3600
```

- The account **must keep its public endpoint reachable**: the browser fetches each PDF directly by its short-lived SAS URL (the case viewer), and Document Intelligence fetches the file by SAS URL too. Access is protected by the SAS signature, and anonymous blob access is disabled. Locking the account to private endpoints would require code changes (stream files through the API; send bytes to Document Intelligence).
- The CORS rule lets the viewer read PDFs from the web app's origin; add the Front Door hostname too if used.
- Consider a lifecycle rule moving blobs older than e.g. 180 days to *Cool* (cheaper storage; the files are immutable).

4.8 Document Intelligence and Azure OpenAI

```
az cognitiveservices account create -g $RG -n $PREFIX-docintel -l $LOC --kind FormRecognizer --sku S0 \
  --custom-domain $PREFIX-docintel
az cognitiveservices account create -g $RG -n $PREFIX-openai -l $LOC --kind AIServices --sku S0 \
  --custom-domain $PREFIX-openai
az cognitiveservices account deployment create -g $RG -n $PREFIX-openai --deployment-name gpt-4.1-mini \
  --model-name gpt-4.1-mini --model-version 2025-04-14 --model-format OpenAI \
  --sku-name GlobalStandard --sku-capacity <free>      # thousands of TPM: the subscription's free capacity
(runbook step 10)
```

- The deployment name must contain `gpt-4.1` (or `gpt-4o`, …) — the app checks it is vision-capable.
- Set the app's caps to **80 %** of the granted quota: `AZURE_OPENAI_MAX_TOKENS_PER_MINUTE` = 0.8 × TPM, `AZURE_OPENAI_MAX_REQUESTS_PER_MINUTE` = 0.8 × RPM (read both from the portal or the `x-ratelimit-limit-*` response headers). Document Intelligence S0 allows 15 requests/s; keep `AZURE_DOCUMENT_INTELLIGENCE_MAX_CALLS_PER_SECOND=10`.
- After the first real traffic, run `python -m scripts.calibrate_llm_tokens` once against the production deployment (costs a few cents) to confirm the token reservations.

4.9 Key Vault and secrets

```
az keyvault create -g $RG -n $PREFIX-kv -l $LOC --enable-rbac-authorization true
for s in jwt-secret owner-db-url app-db-password platform-db-password redis-url storage-conn \
         docintel-key openai-key seed-admin-password; do
  az keyvault secret set --vault-name $PREFIX-kv -n $s --value '<value>'
done
```

Secret	Value
`jwt-secret`	`python -c "import secrets; print(secrets.token_urlsafe(48))"`
`owner-db-url`	`postgresql+psycopg2://fddtowner:<OWNER_PASSWORD>@<pg-host>:5432/fddt?sslmode=require`
`app-db-password` / `platform-db-password`	Two long random passwords (they become the passwords of `fddt_app` / `fddt_platform`)
`redis-url`	`rediss://:<REDIS_KEY>@<redis-host>:10000/0?ssl_cert_reqs=required` (Celery refuses a `rediss://` URL without `ssl_cert_reqs`; URL-encode the key)
`storage-conn`	The storage account connection string (the code signs SAS URLs with the account key)
`docintel-key`, `openai-key`	Resource keys
`seed-admin-password`	The first platform admin's password

URL-encode special characters inside passwords that are embedded in URLs.

4.10 Container Apps environment, identity and shared configuration

```
az containerapp env create -g $RG -n $PREFIX-env -l $LOC \
  --infrastructure-subnet-resource-id $(az network vnet subnet show -g $RG --vnet-name $PREFIX-vnet -n aca
--query id -o tsv) \
  --logs-workspace-id $LA_ID --logs-workspace-key $LA_KEY --enable-workload-profiles
```

Every backend app uses one **user-assigned managed identity** with *AcrPull* on the registry and *Key Vault Secrets User* on the vault. Secrets are referenced as `--secrets jwt=keyvaultref:https://$PREFIX-kv.vault.azure.net/secrets/jwt-secret,identityref:<identity-id> …` and mapped into environment variables with `secretref:`.

Backend environment (identical for API, workers, beat and the migration job):

Variable	Value
`ENVIRONMENT`	`production`
`DATABASE_URL`	`secretref:owner-db-url`
`DATABASE_APP_PASSWORD` / `DATABASE_PLATFORM_PASSWORD`	`secretref:app-db-password` / `secretref:platform-db-password`
`DATABASE_POOLER_HOST` / `DATABASE_POOLER_PORT`	`<pg-host>` / `6432` (tiers M/L). Tier S (Burstable): leave unset and set `DB_POOL_SIZE=3`, `DB_MAX_OVERFLOW=5`
`CELERY_BROKER_URL`, `CELERY_RESULT_BACKEND`	both `secretref:redis-url`
`JWT_SECRET_KEY`	`secretref:jwt-secret`
`AZURE_STORAGE_CONNECTION_STRING`, `AZURE_STORAGE_CONTAINER_NAME`	`secretref:storage-conn`, `documents`
`AZURE_DOCUMENT_INTELLIGENCE_ENDPOINT` / `_KEY`	`https://$PREFIX-docintel.cognitiveservices.azure.com/` / `secretref:docintel-key`
`AZURE_OPENAI_ENDPOINT` / `_KEY` / `_DEPLOYMENT_NAME`	`https://$PREFIX-openai.openai.azure.com/` / `secretref:openai-key` / `gpt-4.1-mini`
`AZURE_OPENAI_MAX_REQUESTS_PER_MINUTE` / `_MAX_TOKENS_PER_MINUTE`	80 % of the deployment's RPM / TPM
`AZURE_DOCUMENT_INTELLIGENCE_MAX_CALLS_PER_SECOND`	`10`
`DEFAULT_MAX_FILE_SIZE_MB` / `DEFAULT_MAX_ZIP_SIZE_MB`	`10` / `300` (limits for new companies; each company's own are set under Platform › Companies)
`QUEUE_ALERT_OLDEST_WAITING_SECONDS`	`300`
`EXTRACTION_WORKER_CONCURRENCY` / `VISION_WORKER_CONCURRENCY`	`4` / `4` (per replica)
`FORENSICS_WORKER_CONCURRENCY`	`0` (one process per vCPU of the container)
`SEED_ADMIN_EMAIL` / `SEED_ADMIN_PASSWORD`	the first platform admin / `secretref:seed-admin-password` — **migration job only**

Full reference: environment-variables.md.

4.11 Database migration job (every release, before the apps)

```
az containerapp job create -g $RG -n $PREFIX-migrate --environment $PREFIX-env \
  --trigger-type Manual --replica-timeout 1800 --replica-retry-limit 0 \
  --image ${PREFIX}acr.azurecr.io/fddt-backend:<version> --cpu 0.5 --memory 1Gi \
  --command sh --args "-c" "alembic upgrade head && python seed.py" \
  --env-vars <the backend variables above> ...
az containerapp job start -g $RG -n $PREFIX-migrate
az containerapp job execution list -g $RG -n $PREFIX-migrate -o table     # wait for Succeeded
```

- **Take a database backup/point-in-time marker before migrating** (Flexible Server keeps automatic backups; note the time so a PITR restore target is known).
- `seed.py` creates the platform admin (`SEED_ADMIN_EMAIL` / `SEED_ADMIN_PASSWORD`) if it does not exist and leaves an existing account unchanged, so it is safe on every release. `python seed.py --reset` restores the seeded password to recover a locked-out admin.
- Do **not** use `start-api.sh` as the API command in production: it migrates and re-seeds on every start, and several API replicas would race.

4.12 Applications

```
IMG=${PREFIX}acr.azurecr.io/fddt-backend:<version>
# API — internal ingress only
az containerapp create -g $RG -n fddt-api --environment $PREFIX-env --image $IMG \
  --ingress internal --target-port 8000 --cpu 1 --memory 2Gi --min-replicas 2 --max-replicas 4 \
  --command uvicorn --args app.main:app --host 0.0.0.0 --port 8000 --workers 2 --proxy-headers --forwarded-
allow-ips '*' \
  --env-vars <backend variables>
# Workers — no ingress
az containerapp create -g $RG -n fddt-worker-extraction --environment $PREFIX-env --image $IMG \
  --cpu 1 --memory 2Gi --min-replicas 1 --max-replicas 1 --command sh --args start-worker.sh extraction --
env-vars <…>
az containerapp create -g $RG -n fddt-worker-vision --environment $PREFIX-env --image $IMG \
  --cpu 1 --memory 2Gi --min-replicas 1 --max-replicas 1 --command sh --args start-worker.sh vision --env-
vars <…>
az containerapp create -g $RG -n fddt-worker-forensics --environment $PREFIX-env --image $IMG \
  --cpu 2 --memory 4Gi --min-replicas 0 --max-replicas 4 --command sh --args start-worker.sh forensics --
env-vars <…>
# (scale-to-zero needs the KEDA Redis rules on forensics_queue — see the runbook, step 15.2)
# Beat — exactly one
az containerapp create -g $RG -n fddt-beat --environment $PREFIX-env --image $IMG \
  --cpu 0.5 --memory 1Gi --min-replicas 1 --max-replicas 1 --command sh --args start-beat.sh --env-vars <…>
# Web — the only public app
az containerapp create -g $RG -n fddt-web --environment $PREFIX-env \
  --image ${PREFIX}acr.azurecr.io/fddt-web:<version> --ingress external --target-port 80 \
  --cpu 0.25 --memory 0.5Gi --min-replicas 2 --max-replicas 3 \
  --env-vars API_UPSTREAM=https://fddt-api.internal.<environment-default-domain>
```

(Add `--registry-server` , `--registry-identity` and `--user-assigned` for the managed identity on each command; the replica counts above are tier M — take them from section 2. The CLI parses `--host` -style tokens inside `--args` as its own flags, so in practice create the apps from YAML as the runbook does.)

- `API_UPSTREAM` is the API's **internal FQDN** (`az containerapp show -n fddt-api --query properties.configuration.ingress.fqdn`). nginx strips `/api` , forwards to it, and refuses bodies over `UPLOAD_BODY_LIMIT` (default **11m**) — or, for `/api/bulk-uploads` (streamed, unbuffered), `BULK_UPLOAD_BODY_LIMIT` (default **301m**) — before they reach Uvicorn. Upload limits are per company, so these

two must be at least the **largest** per-company limit granted (plus ~1 MB of multipart overhead for single files): when a platform admin gives a company 20 MB / 500 MB, redeploy `fddt-web` with `--env-vars UPLOAD_BODY_LIMIT=21m BULK_UPLOAD_BODY_LIMIT=501m`.

- Workers need no probes beyond the default; give the API a liveness probe on `GET /health`.
- Rolling update order for a release: build images → run `fddt-migrate` → update `fddt-api` → update the workers and beat → update `fddt-web`. Migrations are additive within a release, so old workers keep working during the roll.
- **Restart the workers whenever their code changes** (a new revision does this).

4.13 Domain and TLS

```
az containerapp hostname add -g $RG -n fddt-web --hostname app.<client-domain>
az containerapp hostname bind -g $RG -n fddt-web --hostname app.<client-domain> \
  --environment $PREFIX-env --validation-method CNAME     # free managed certificate
```

Or put Front Door Standard in front (managed certificate on Front Door, origin = `fddt-web`) and restrict `fddt-web` ingress to the Front Door service tag. Add the final origin to the Blob CORS rule (4.7).

4.14 First login and smoke test

1. Sign in as the seeded platform admin; change nothing else until the checks below pass.
2. **Platform › Companies** → create the client company (it receives the 38 template rules and 30/60 thresholds; its issuer registry is empty). **Platform › Users** → create its reviewers and users.
3. As a company user, upload one sample PDF; confirm it reaches a risk tier, the PDF renders in the case viewer (Blob CORS works) and **Platform › Processing queues** shows all three queues with consumers.
4. Upload an image and a password-protected PDF — both must be refused with their specific messages.
5. Generate a report as a reviewer and open it.
6. As the company's Reviewer L2, load the company's real **issuer registry** (until then almost every document flags `issuer.not_in_registry`).

5. Operations

Monitoring and alerts

Alert	Query / metric	Suggested threshold
Queue backlog	`ContainerAppConsoleLogs_CL | where Log_s has "queue_alert"`	any row in 5 min (the app logs it when the oldest waiting task > 300 s)
Worker errors	`ContainerAppConsoleLogs_CL | where ContainerAppName_s startswith "fddt-worker" and Log_s has_any ("ERROR", "Traceback")`	> 5 in 15 min
Database permission errors	`… Log_s has "permission denied"`	any (a write needs a grant the app role lacks)
Azure throttling	`… Log_s has "429"`	> 10 in 15 min — raise TPM or lower the caps
Stuck documents re-queued	`… Log_s has "stuck_documents_requeued"`	any — Redis lost queued tasks (restart/outage); the documents were queued again automatically

Alert	Query / metric	Suggested threshold
Sign-in lockouts	`… Log_s has "login_lockout"`	> 10 in 15 min (password guessing)
Failed documents	`audit_log` events `document_processing_failed` / `*_failed` (query through the platform audit log)	review daily
PostgreSQL	CPU > 80 %, storage > 80 %, `client_connections_waiting` (PgBouncer) > 0 for 5 min	
Redis	Used memory > 70 %, server load > 80 %	
API	`fddt-api` 5xx rate, replica restarts	

`queue_metrics {json}` lines (one per queue per minute) can be charted directly in Log Analytics (`parse_json(extract("queue_metrics (.*)", 1, Log_s))`).

Backups and disaster recovery

Data	Protection	Restore
PostgreSQL	Automatic backups, 35-day point-in-time restore; zone-redundant HA (L; M once enabled) fails over automatically; without HA a zone outage means a restore	PITR to a new server, repoint `DATABASE_URL` / pooler host
Blob Storage	ZRS; enable **soft delete** (14 days) and **versioning** for defence in depth (the app never deletes or overwrites)	Undelete / previous version
Redis	Holds only queued tasks, fair-share counters, sign-in throttle and rate-limiter windows	A lost broker loses queued tasks; the stuck-document job queues the affected documents again within ~15 minutes (checks of documents whose extraction had already finished are not re-run)
Configuration	Key Vault (soft delete + purge protection on) and the image tags	Redeploy

Recovery objectives with tier M as designed: RPO ≈ 5 minutes (PostgreSQL log backups), RTO ≈ 1 hour (restore + repoint), excluding in-flight pipeline tasks.

Routine tasks

- **Secret rotation:** rotate storage/Document Intelligence/OpenAI keys with the two-key pattern; rotating `JWT_SECRET_KEY` signs everyone out. To rotate `fddt_app` / `fddt_platform`, `ALTER ROLE … PASSWORD` as the owner, then update the Key Vault secrets and restart the apps.
- **Usage reconciliation** runs nightly (02:00 UTC); **Billing & usage** feeds client invoicing.
- **Re-calibrate token reservations** (`scripts.calibrate_llm_tokens`) after any model, prompt or render change; re-check the 80 % caps after any quota change.
- **Schema docs:** regenerate `docs/database/*.md` with `docs/tools/gen_db_docs.py` after each migration.

6. Go-live checklist

Must do

- [ ] `JWT_SECRET_KEY`, both app-role passwords, the owner password and `SEED_ADMIN_PASSWORD` are long random values in Key Vault; nothing secret is in an image or the repo.
- [ ] The migration job ran (`alembic upgrade head` → head `c9e3a1b5d7f4` or later) and the seed step was removed or pinned to the real password.
- [ ] `DATABASE_URL` ends in `?sslmode=require`; tiers M/L connect through the built-in PgBouncer (6432).
- [ ] Redis is Non-Clustered and both Celery URLs use `/0` over `rediss://`.
- [ ] gpt-4.1-mini Global Standard deployment uses the free capacity (no quota request); the app caps at 80 % of it.
- [ ] Exactly one `fddt-beat` replica (beat + the `housekeeping_queue` worker).
- [ ] HA decision for PostgreSQL and Redis recorded with the client (tier M launches without HA).
- [ ] Blob container private, anonymous access off, CORS limited to the app origin, soft delete on.
- [ ] Only `fddt-web` has public ingress.
- [ ] Alerts from section 5 created and routed to an on-call address.
- [ ] Data-residency decision for the Global Azure OpenAI deployment recorded with the client.
- [ ] The client's issuer registry loaded; risk weights reviewed with the client.
- [ ] A full browser walkthrough of every role (submitter, Reviewer L1, Reviewer L2, platform admin).

Known gaps to accept or schedule (from azure-migration-notes.md)

- No forgot-password flow (users change their own password; admins reset others), tokens in `localStorage`. Sign-in throttling exists (per address and per IP).
- No malware scan and no page-count limit on uploads (type, size, corruption and password are checked).
- No automatic retry of a failed Azure call. (Documents whose queued tasks were lost are re-queued by the stuck-document job.)
- Keys and connection strings, not managed identity, for Storage / Document Intelligence / OpenAI.

Sources

- Azure Retail Prices API — `https://prices.azure.com/api/retail/prices` (UAE North and West Europe, retrieved 2026-10-03)
- Billing in Azure Container Apps
- PgBouncer in Azure Database for PostgreSQL Flexible Server
- Azure Cache for Redis retirement FAQ
- Azure Managed Redis architecture (clustering policies)
- Azure Managed Redis pricing
- Azure Front Door pricing